HBS 9-119-020 · Datar & Bowler (2018)

LendingClub (A)

Data Analytic Thinking

How Emily Figel built a default-prediction model on 413,000 peer-to-peer loans — and what almost went wrong.

Presented by

Kishore Kunal

Roll No. 38BM21 · PGDM-BM & PGDM-BA · MDI Gurgaon

Course

Strategy Formulation & Implementation

Prof. Sangeeta Shah Bharadwaj · Term IV (Jun–Aug 2026)

ROADMAP FOR TODAY

Agenda

01 **The Setup**

Emily Figel, wealth manager–in–waiting, meets peer-to-peer lending

02 **The Business**

LendingClub's model, mechanics, grades, and the $10k example

03 **The Data**

413,000 loans, 100+ features, and one messy target variable

04 **The Split**

Training vs. holdout, five-fold cross-validation, balanced or not

05 **The Models**

First attempt (too perfect), leakage found, second attempt (realistic)

06 **The Risks**

Borrower quality, transparency, and LendingClub as counterparty

07 **Insights**

What Figel — and we — should take away before we invest a dollar

Meet Emily Figel — and her first client-facing question

THE PROTAGONIST

Emily Figel

- MBA student, joining a Chicago wealth-management firm in the fall
- Fascinated by alternative banking models
- Advises aggressive clients — small mistakes scale fast

HER FIRST PROJECT

Is LendingClub a reasonable investment vehicle for the firm's aggressive clients?

HER PLAN — WHAT SHE ACTUALLY WANTS TO DO

1. **Learn the basics** — of predictive modelling using LendingClub's own historical data
2. **Get a feel** — for the opportunities and risks of investing on the platform
3. **Build a classifier** — to predict whether a new borrower will default or repay
4. **Use predictions** — as one input among several before funding any loan

LendingClub at a glance

2007

Platform launched — one of the first P2P lenders alongside Zopa (UK), Prosper (US)

Dec 2014

Largest tech IPO of the year — raised nearly $900M; poster child for 'fintech'

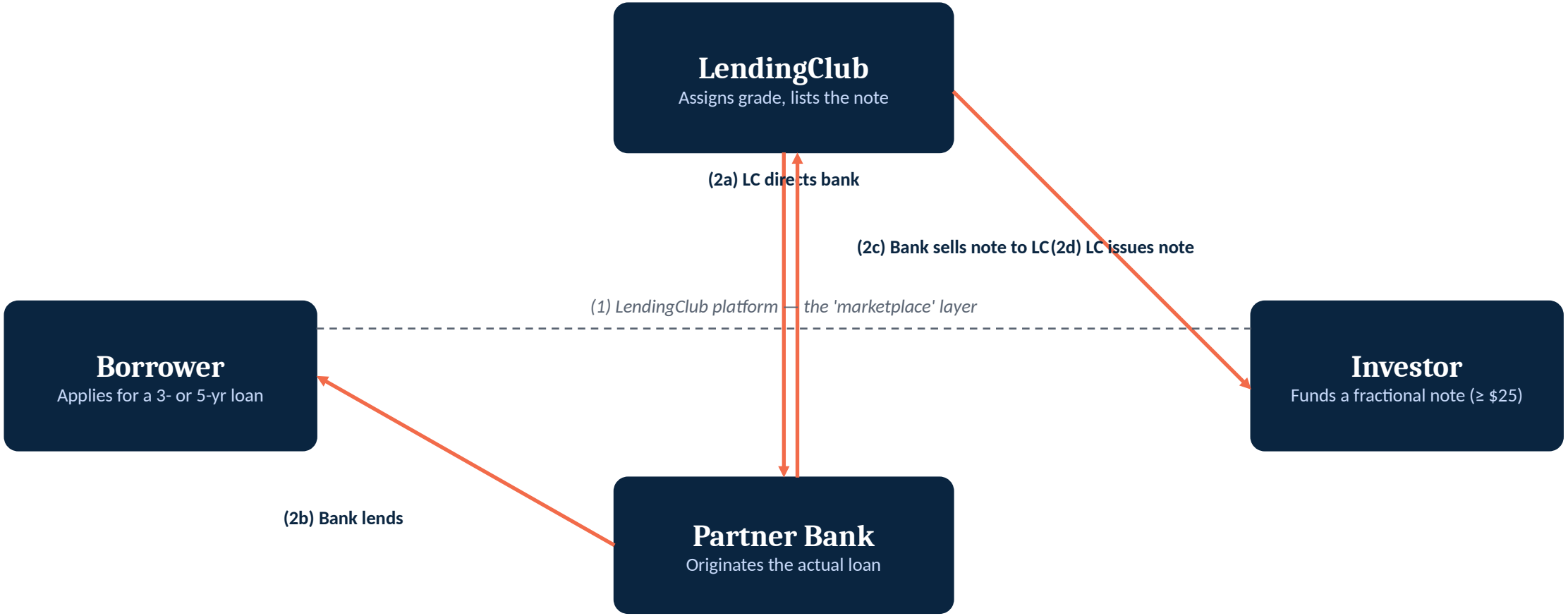
$1.5B / qtr

Loans issued by 2016 — up from $1B/qtr at IPO and $1B cumulative in 2012

THE THREE-SIDED PROPOSITION

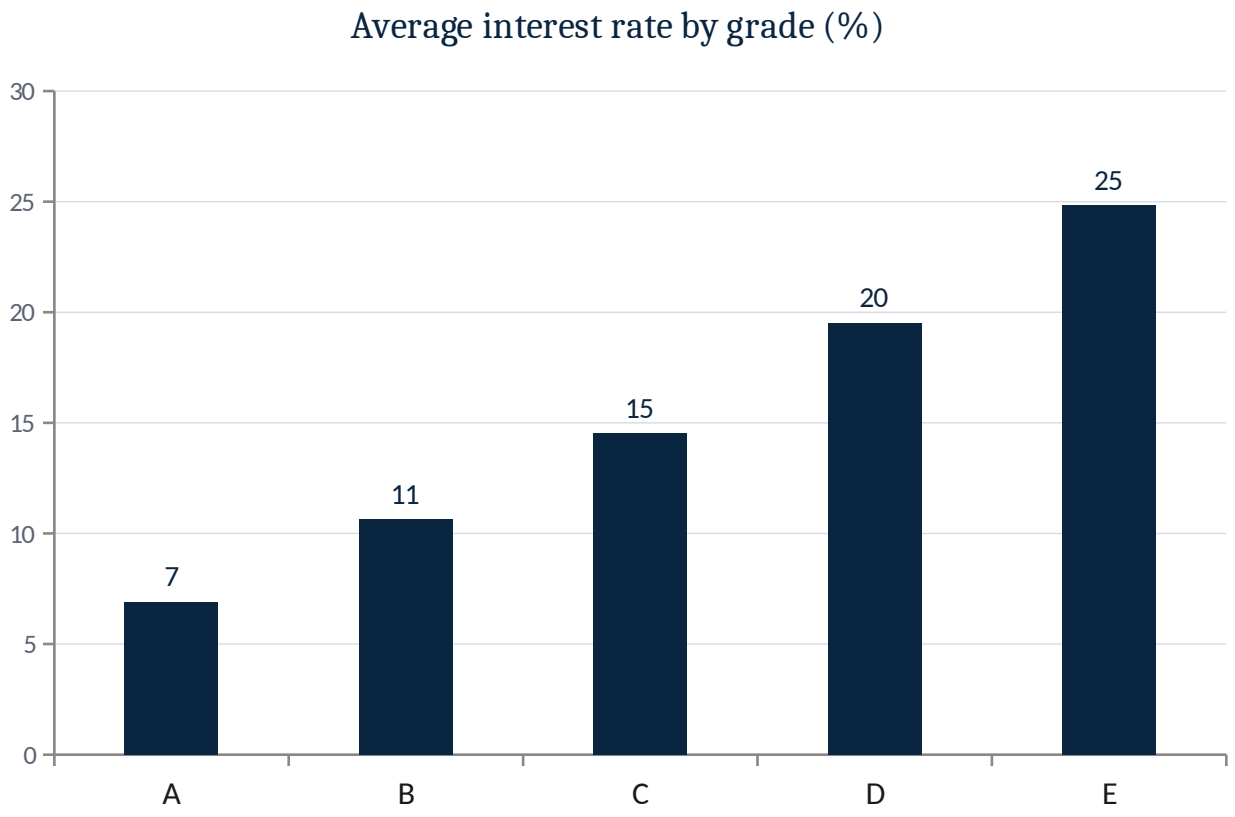
Borrowers	3- or 5-year unsecured personal loans; FICO ≥ 660; most used it for debt consolidation, credit-card refinance, home renovation
Investors	Individuals (and later institutions) fund fractional 'notes' (min $25) at the interest rate LC assigns; earn interest, take default risk
LendingClub	Runs the online platform, assigns grades, charges 3.5% origination + 1% servicing; ~15% of 2016 loans still from small individual investors

The lending process is not really 'peer-to-peer'



Why this matters: The investor never lends directly to the borrower — she buys a note from LendingClub. If LC files for bankruptcy, she becomes an unsecured creditor with no guaranteed claim on her borrower's payments.

Grades set the price — 35 sub-grades from 6% to 27%



Interest rate by sub-grade (%)

Sub	A	B	C	D	E
1	6.03	9.43	13.06	17.47	22.90
2	6.14	10.07	13.58	18.45	23.87
3	6.83	10.56	14.52	19.42	24.84
4	7.46	11.05	15.49	20.39	25.81
5	8.08	12.13	16.01	21.85	26.77

Read the price like a bond: A1 borrowers pay 6.03% — barely above Treasuries. E5 borrowers pay 26.77% — credit-card territory. Grade and sub-grade are two of **100+ features** LendingClub publishes with every loan.

Follow the money — a $10,000 A1 loan at 6.03%

$10,000
Loan amount
Borrower applies, LC grades A1

→

3.5%
Origination fee
LC deducts $350 up front

→

$9,650
Actually disbursed
What lands in borrower's account

→

$304.36
Monthly payment
36 months, of which $3.05 goes to LC as 1% servicing

→

$10,956.79
Total repaid
By end of month 36 — $956.79 total interest

EFFECTIVE INTEREST RATE ≠ STATED RATE

6.24%

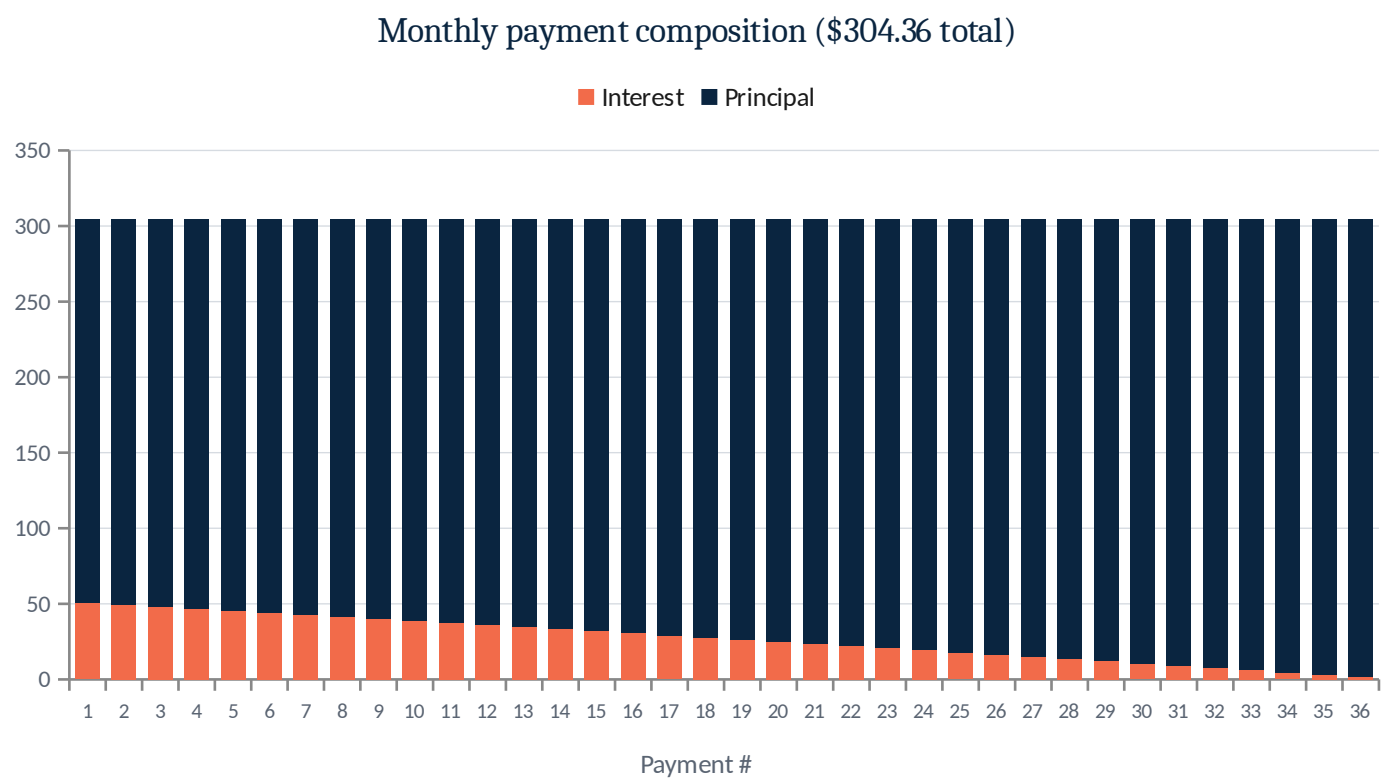
effective rate on cash actually received

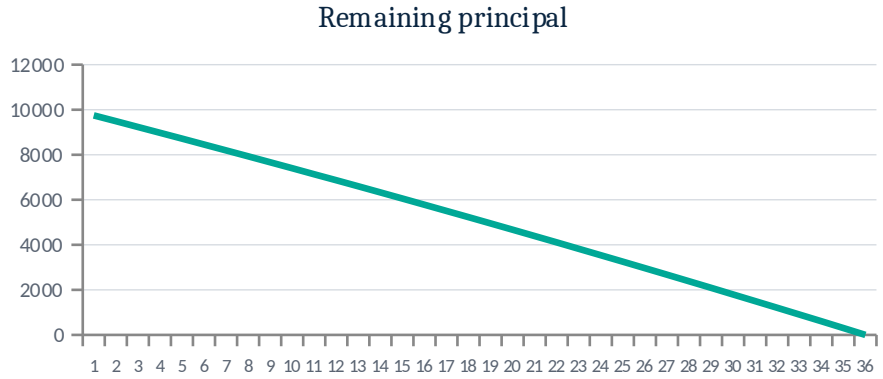
- Interest paid in Year 1 **$10,000 × 0.0603 = $603**
- Divided by amount disbursed **$603 / $9,650 = 6.24%**

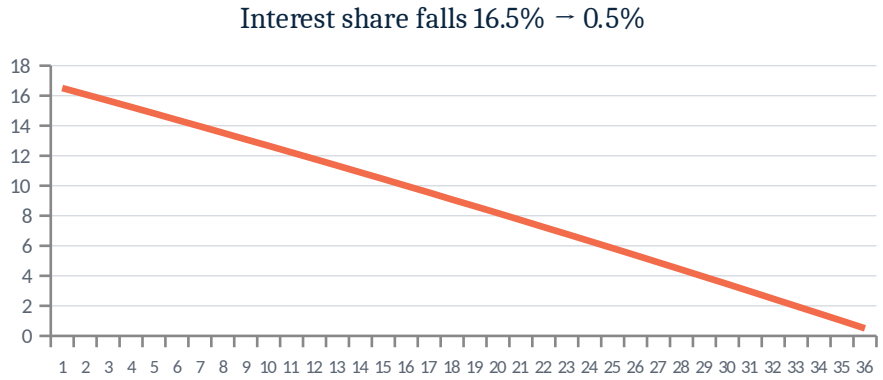
WHO EARNS WHAT ON THIS LOAN

- **LendingClub** $350 origination + 1% servicing on every payment
- **Investor** $304.36 − $3.05 = $301.31 per month × 36 months
- **Borrower** Pays 6.03% stated / 6.24% effective on cash received
- **If default** LC charges investors a collection fee to recover principal

Interest is front-loaded — like a mortgage in miniature







The default-timing insight: Because interest is paid first, an investor who suffers a default late in the loan has already received most of her return. That's why Table 2 (17.97% default rate) and Table 3 (7.39% net return) both make sense — defaults happen ***after*** borrowers have already made real payments.

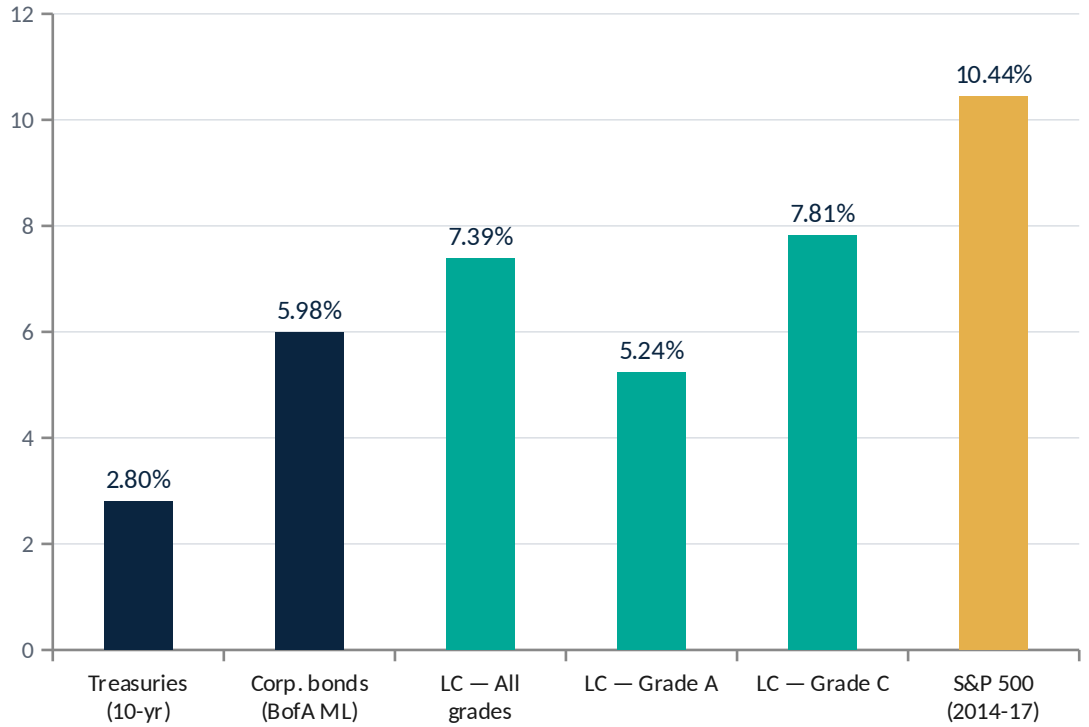
How LendingClub compares as an asset class

WHAT THE INVESTOR ACTUALLY SEES

- **Sub-grade & interest rate** (e.g. A3 · 6.83%)
- **Term** (36 or 60 months) and FICO band
- **Loan amount** (e.g. $8,000)
- **Purpose** (debt refi, credit-card, small biz, other)
- **% funded** and time remaining to fund
- **Minimum ticket** $25 → build a diversified portfolio of fractional notes

Figel's specific goal: *build a portfolio whose returns are commensurate with the risks she takes — i.e. beat corporate bonds without taking equity-like drawdowns.*

Annualized returns — 2017 snapshot (%)



LendingClub sits between corporate bonds and equities — but Figel still has to pick the right grade mix.

The data — 413,000 loans, 100+ features, one messy label

3
CSV files
2013 → Q1 2015, merged into a single table

413K
records
one row per loan application

100+
features
borrower, credit history, loan terms, LC actions

1
target
loan_status — collapsed to default (1) / repay (0)

WHY THIS DATA EXISTS AT ALL

Transparency was strategic. LendingClub anonymised its historical loan data and made it downloadable from its own website — not just new opportunities, but every loan since 2007. That's how it built the trust investors demanded in exchange for their capital. It's also how Figel could build a model at all.

ACADEMIC / STATISTICAL

Include a feature only if you have a reasonable causal hypothesis. Anything else is 'data mining.' Explanation > prediction.

DATA SCIENCE / PREDICTIVE

"Causality is elusive. Data scientists have it easier with predictive analytics — it just needs to work. Prediction trumps explanation."
— Eric Siegel

What's inside those 100+ features (Exhibit 6)

APPLICATION

- id, zip_code, addr_state
- home_ownership, emp_title, emp_length
- annual_inc (± joint), mort_acc
- desc, purpose, application_type
- loan_ammt, installment, int_rate, term
- verification_status (± joint)

CREDIT PROFILE

- dti — debt-to-income ratio
- fico_range_high / _low
- earliest_cr_line, mo_sin_old_rev_tl_op
- total_acc, tot_cur_bal, avg_cur_bal
- total_bc_limit, tot_hi_cred_lim
- percent_bc_gt_75
- inq_last_12m, inq_last_6mths

ACTIVE ACCOUNTS

- num_sats — satisfactory accounts
- revol_bal, revol_util
- delinq_2yrs, delinq_amnt
- mths_since_last_delinq
- num_accts_ever_120_pd
- num_tl_30dpd / 90g_dpd_24m / 120dpd_2m
- tax_liens, pub_rec, pub_rec_bankruptcies

LOAN & PAYMENTS

- funded_amnt
- last_fico_range_high / _low
- last_credit_pull_d
- loan status, total_pymnt, total_rec_prncp
- total_rec_int, last_pymnt_amnt / _d
- out_prncp, out_prncp_inv, pymnt_plan

HARDSHIP

- hardship_flag, hardship_type
- hardship_reason, hardship_status
- hardship_amount

SETTLEMENT

- debt_settlement_flag
- debt_settlement_flag_date
- settlement_percentage

Red-shaded groups are the ones Figel later flags as leakage-prone — they contain information that only exists AFTER the loan's outcome is known.

Defining the target — six raw classes, two Figel keeps

Value	Definition	Target label	Figel's decision
Fully paid	The loan has been repaid in full.	**0**	**Repay**
Current	The loan is still being repaid.	**—**	**Excluded**
Late 16-30 days	Not current for 16-30 days.	**—**	**Excluded**
Late 31-120 days	Not current for 31-120 days.	**—**	**Excluded**
Default	Not paid in 121 days or more.	**1**	**Default**
Charged Off	In default 30+ days, no more payments expected.	**1**	**Default**

WHY EXCLUDE 'CURRENT' AND LATE?

Their outcomes aren't known yet.

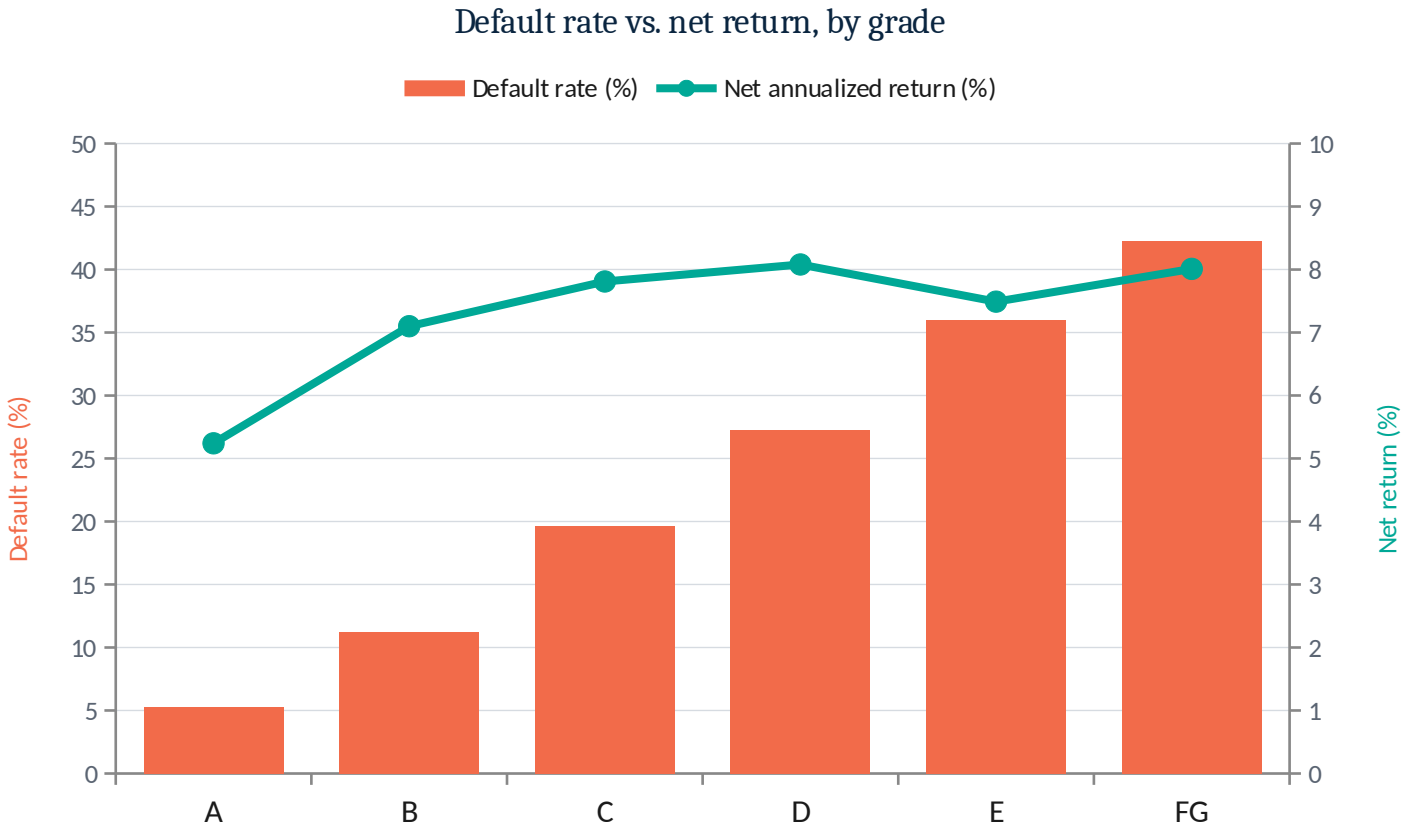
A 'Current' loan might repay in full — or default next quarter.

A 'Late' loan might cure — or worsen into Charged Off.

Labelling them either way would inject noise into the target.

Final labelled dataset: 413,000 loans · **75,000 defaulters (1)** · **338,000 repayers (0)** → *~18% base default rate*

The paradox — high default rates, still positive returns



HOW TO READ IT

Default rate climbs sharply — from **5.3% (A) to 42.2% (FG).**

Net return is nearly flat — **5.2% to 8.1%.**

Overall default rate = 17.97%, net return = 7.39%.

Why? Because 'default' is discrete, but payment is continuous. Even defaulted borrowers pay interest and principal for months before defaulting.

Investable implication: A model that only predicts default/repay is not enough. Figel really wants to predict **expected return conditional on grade + features** — but default classification is the tractable first step.

Splitting 413K loans — training, cross-validation, holdout

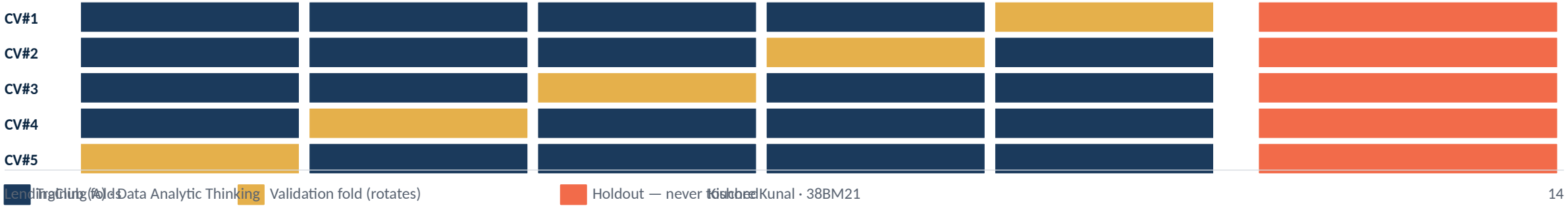
STEP 1 · 80 / 20 RANDOM SPLIT

TRAINING SET · 80% · 330,000 loans (60K defaulters + 270K repayers) | HOLDOUT · 20% · 83K

STEP 2 · TRAINING SET → 5 EQUAL FOLDS

Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	HOLDOUT
66,000 loans	66,000 loans	66,000 loans	66,000 loans	66,000 loans	83,000
12K def + 54K rep	12K def + 54K rep	12K def + 54K rep	12K def + 54K rep	12K def + 54K rep	*Untouched until final test*

STEP 3 · IN EACH CV RUN, ONE FOLD IS THE VALIDATION SET



Five steps to a validated model (Exhibit 7)

1

Build 5 candidate models

Try logistic regression, decision trees, gradient boosting and variants on the same folds — 5 model families in parallel

2

Cross-validate each

For every family, run CV #1-5 — train on 4 folds, score on the 5th — and average the 5 CV scores

3

Pick the winner

Select the model type with the highest average CV score (best accuracy on unseen data)

4

Re-train on all 80%

Take the chosen model type, tune its hyper-parameters, and train it once on the entire training set

5

Test on holdout

Score the final model on the 20% holdout it has never seen — this is the honest performance estimate

> *"Since the holdout set was not used to create the model, there's no way the model could have captured its esoteric aspects, its eccentricities."*
>
> — Eric Siegel

Balanced or unbalanced? Figel's genuine dilemma

Realistic

USE ALL 264,000 OBSERVATIONS

PROS

- ✓ Reflects the real base rate of default (18%)
- ✓ No information thrown away
- ✓ Model's probabilities are directly interpretable

CONS

- × Model has 4.5× more repayers than defaulters
- × Predictions can drift toward the majority class
- × Rare-class recall (default) may suffer

Focused

BALANCED · 48K DEFAULT + 48K REPAY

PROS

- ✓ Model 'sees' defaulters as often as repayers
- ✓ Better sensitivity to the risk signal
- ✓ Focus of what Figel actually cares about

CONS

- × No longer representative of the population
- × Probability outputs need recalibration
- × Introduces a matching decision for validation folds

And if she balances the training set: does she test on a balanced 12K + 12K fold, or the full 66K fold that reflects reality? The case leaves this open — but the honest answer is the imbalanced fold, because that's how production loans will actually arrive.

First attempt — a suspiciously perfect model

WHAT FIGEL SAW

Regularized logistic regression trained on all 264K training observations.

Log-losses across all 5 CV folds were ≈ 0.

Every defaulter was scored 1.00. Every repayer was scored 0.00.

FIGEL'S HUNCH

"Doubt quickly crowded out my initial sense of awe."

In a real classification problem, near-perfect accuracy almost always means one of two things:

(1) the task is trivial — usually not, or

(2) a feature is leaking the label.

EXHIBIT 8 · MODEL SCORING — ATTEMPT 1

Rec #	Actual (1 = bad)	Attempt 1
1	1	1.00
2	1	1.00
3	1	1.00
4	1	1.00
5	1	1.00
6	1	1.00
7	1	1.00
8	1	1.00
9	1	1.00
10	1	1.00
11	0	0.00
12	0	0.00
13	0	0.00
14	0	0.00
15	0	0.00
16	0	0.00
17	0	0.00
18	0	0.00
19	0	0.00

The leak — 20 features that only exist after the outcome

OUTCOME-COUPLED PAYMENT DATA

- loan status
- total_pymnt · total_rec_prncp · total_rec_int
- last_pymnt_amnt · last_pymnt_d
- out_prncp · out_prncp_inv
- pymnt_plan

POST-ORIGINATION CREDIT UPDATES

- last_fico_range_high / _low
- last_credit_pull_d

HARDSHIP & SETTLEMENT PLANS

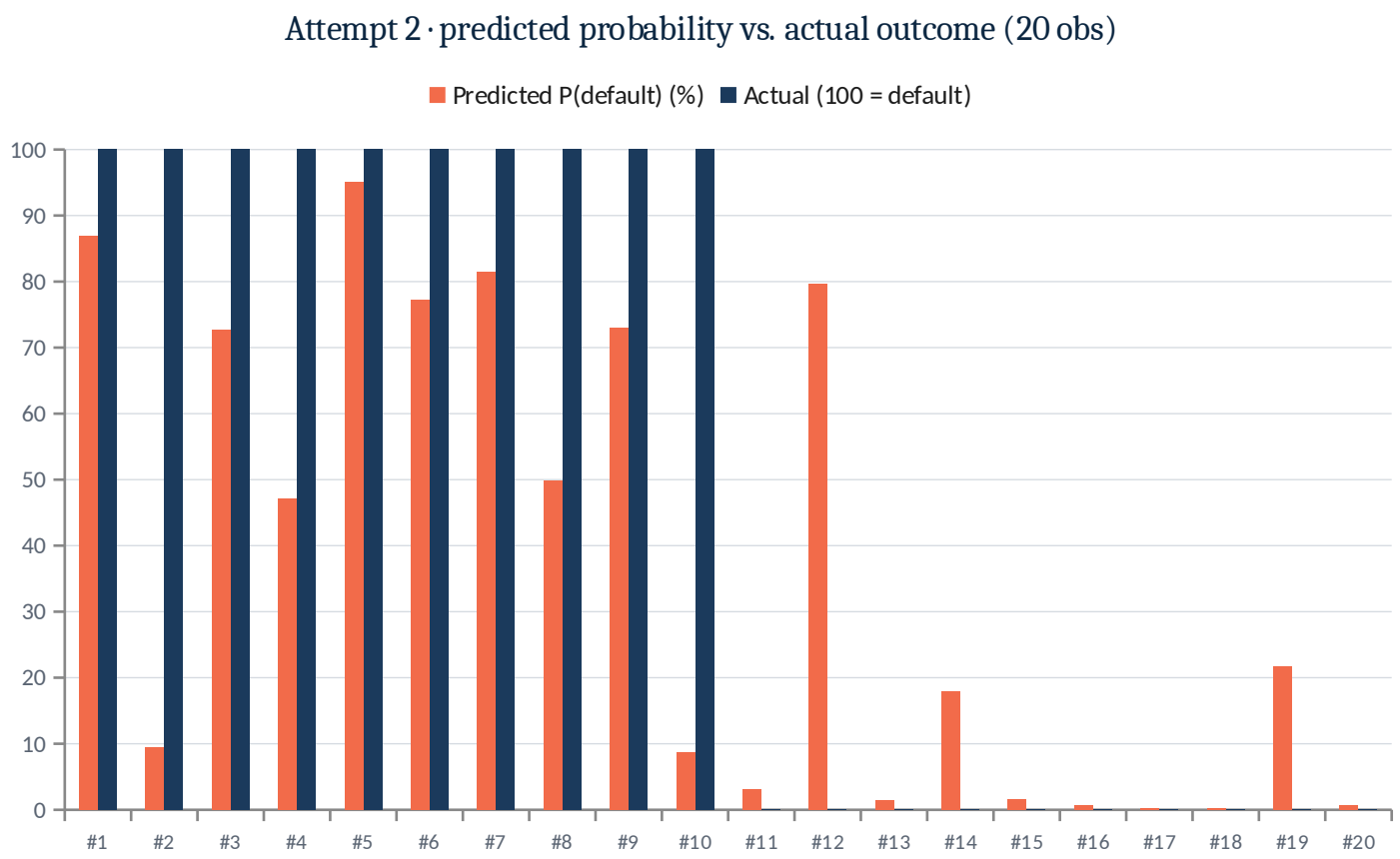
- hardship_flag · hardship_type · hardship_reason
- hardship_status · hardship_amount
- debt_settlement_flag (+ date)
- settlement_percentage

FUNDING

- funded_amnt — nearly identical to loan_amnt for successful funds

The common thread: each feature is only populated once the loan has run for a while — so by construction it 'knows' the outcome. Data leakage in a nutshell.

Second attempt — the messy, believable result



HOW TO READ IT

The model is now **directionally right,** not oracle-right.

Wins: 8 of 10 actual defaulters have P > 47%; 8 of 10 actual repayers have P < 22%.

Misses: Record 2 (default, P = 9%) and Record 12 (repay, P = 80%) show the model does make mistakes at both extremes.

That's what a good risk model looks like.

Next design choice: where to set the cut-off? A 50% threshold minimises misclassification, but Figel's real optimand is expected return — so she may accept some false positives to avoid the worst false negatives.

Attempt 1 vs. Attempt 2 — the same 20 borrowers

Rec	Actual	Attempt 1	Attempt 2	Direction
1	1	1.00	86.96%	✓
2	1	1.00	9.47%	✗
3	1	1.00	72.75%	✓
4	1	1.00	47.14%	✗
5	1	1.00	95.05%	✓
6	1	1.00	77.27%	✓
7	1	1.00	81.48%	✓
8	1	1.00	49.76%	✗
9	1	1.00	72.95%	✓
10	1	1.00	8.64%	✗
11	0	0.00	3.07%	✓
12	0	0.00	79.58%	✗
13	0	0.00	1.45%	✓
14	0	0.00	17.90%	✓
15	0	0.00	1.55%	✓
16	0	0.00	0.75%	✓
17	0	0.00	0.15%	✓
18	0	0.00	0.20%	✓

ATTEMPT 1 · WITH LEAKAGE

20 / 20

correct — probabilities collapse to 0 or 1

Log-loss ≈ 0 across all folds. Not skill — data leakage.

ATTEMPT 2 · LEAKAGE REMOVED

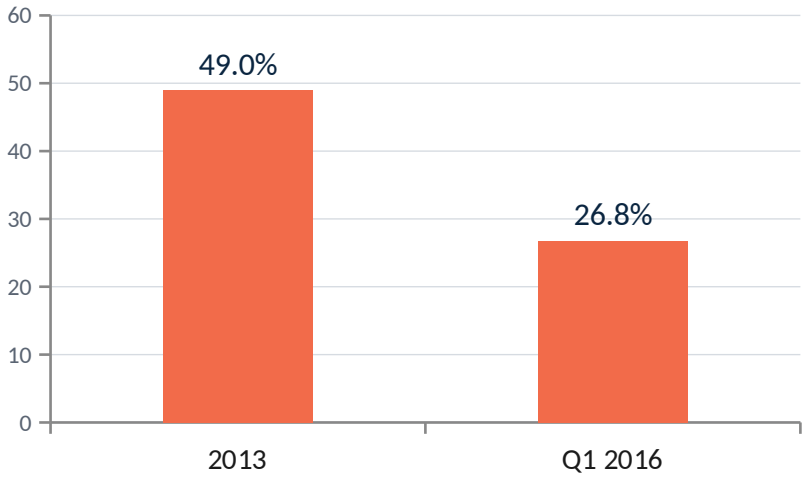
16 / 20

directionally correct at a 50% cut-off

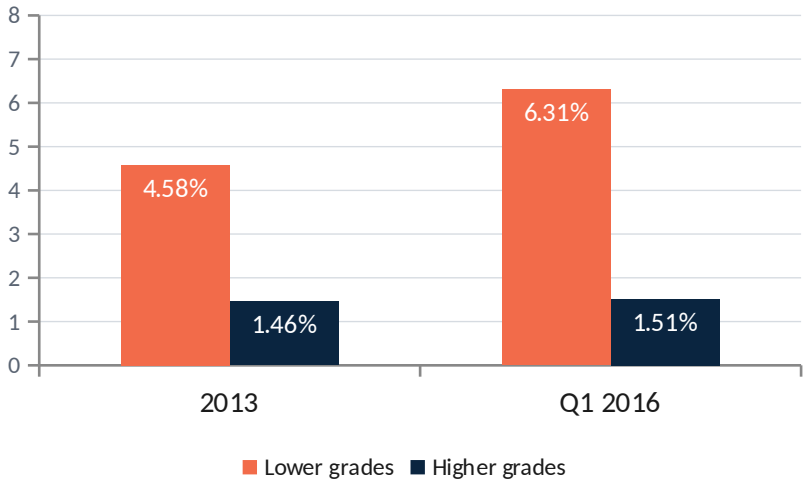
Probabilities now spread realistically — the model is confident when features are decisive, uncertain when they're not.

Risk 1 — borrower quality is drifting

Income verification collapsed



Charge-offs by grade (%)



Even so — LC delinquency rates stayed broadly in line with US consumer-loan rates tracked by the Federal Reserve (1.98% – 2.56% from Q1 2013 to Q1 2018). The concern is the trend, not the level.

THE HEADLINE FROM THE WSJ

+38%

rise in loan charge-offs between 2013 and 2016

AND FROM RESEARCH ON BORROWERS

Nearly half of 'debt consolidation' borrowers carried **10% more credit-card debt** 12 months after taking the loan.

LendingClub loans became a stop-gap for overspending, not a fix for it.

Risk 2 — transparency is the whole business model

WHY TRANSPARENCY IS LOAD-BEARING

- Investors assess loans by reading LendingClub-published data — grade, sub-grade, FICO band, DTI, purpose.
- If those data points are wrong, every investor's risk-return calculation is wrong.
- Institutional investors (Jefferies, hedge funds, endowments) signalled the data was accurate — which is why the small individual investors trusted it too.
- Break that trust once, and the whole two-sided marketplace unwinds.

THE 2016 JEFFERIES INCIDENT

1. **Jefferies buys loans**
   The bank agrees to purchase loans of a specific grade level from LendingClub.
2. **$22M mislabelled**
   After the transaction, Jefferies discovers at least $22M of the loans did NOT meet the specified grade.
3. **It was known internally**
   Several LendingClub executives had known about the mistake — and said nothing.
4. **Fallout**
   Founder & CEO Renaud Laplanche resigns. Executives fired. LC 'redoubles efforts' on data-integrity controls.

Risk 3 — LendingClub itself is a counterparty

THE INVESTOR THINKS SHE OWNS…

A note representing a specific borrower's loan.

So if 'her' borrower repays every month for 3 years, she expects 36 monthly payments of ≈$304 minus servicing.

That's the 'peer-to-peer' promise.

WHAT SHE ACTUALLY HOLDS IF LC FILES BANKRUPTCY

An unsecured claim against LendingClub — not the borrower.

Even if her borrower keeps paying, the cash flows into the LC estate first. She joins the queue of unsecured creditors and takes whatever the bankruptcy process leaves.

There's no guarantee she gets anything back.

Investor takeaway: the model tells her about borrower risk. It says nothing about LendingClub-the-company risk. Both need to be sized before she puts client money on the platform.

Six things to take away from Case A

01 The label is a decision, not a given

Fully-Paid + Charged-Off = supervised task. 'Current' and 'Late' are excluded because their outcomes aren't observed yet.

02 Class imbalance is a design choice

18% default rate is the truth — balancing helps the model see defaults but you must recalibrate probabilities before you act.

03 Cross-validation is the honest way to choose

Five folds keep model selection out of the holdout. The holdout answers only one question: will this generalise?

04 Perfect accuracy = suspect

Attempt 1 wasn't skill; it was leakage from post-outcome features (payments, hardship, last-FICO).

05 Default ≠ Loss

Table 2 vs Table 3: 18% defaults but 7.4% net return. Borrowers pay interest and principal for months before defaulting.

06 Model risk is only half the story

Even a well-tuned classifier can't tell Figel about income-verification drift, Jefferies-style data mislabelling, or LC's own solvency.

PART TWO

Case B

Decision Trees & Random Forests

From a good-enough regression to models that a decision-maker can actually read — and one that wins on log loss.

HBS 9-119-021

Why decision trees? Because rules beat coefficients

INDUSTRY ANALYST DEAN ABBOT

> *"All trees can be read as a series of **'if-then-else' rules** that ultimately generate a predicted value — much more accessible to decision-makers than mathematical formulas."*

1 — Human-readable
A tree is a nested set of if-then rules. A loan officer, a lawyer, and a regulator can all follow it.

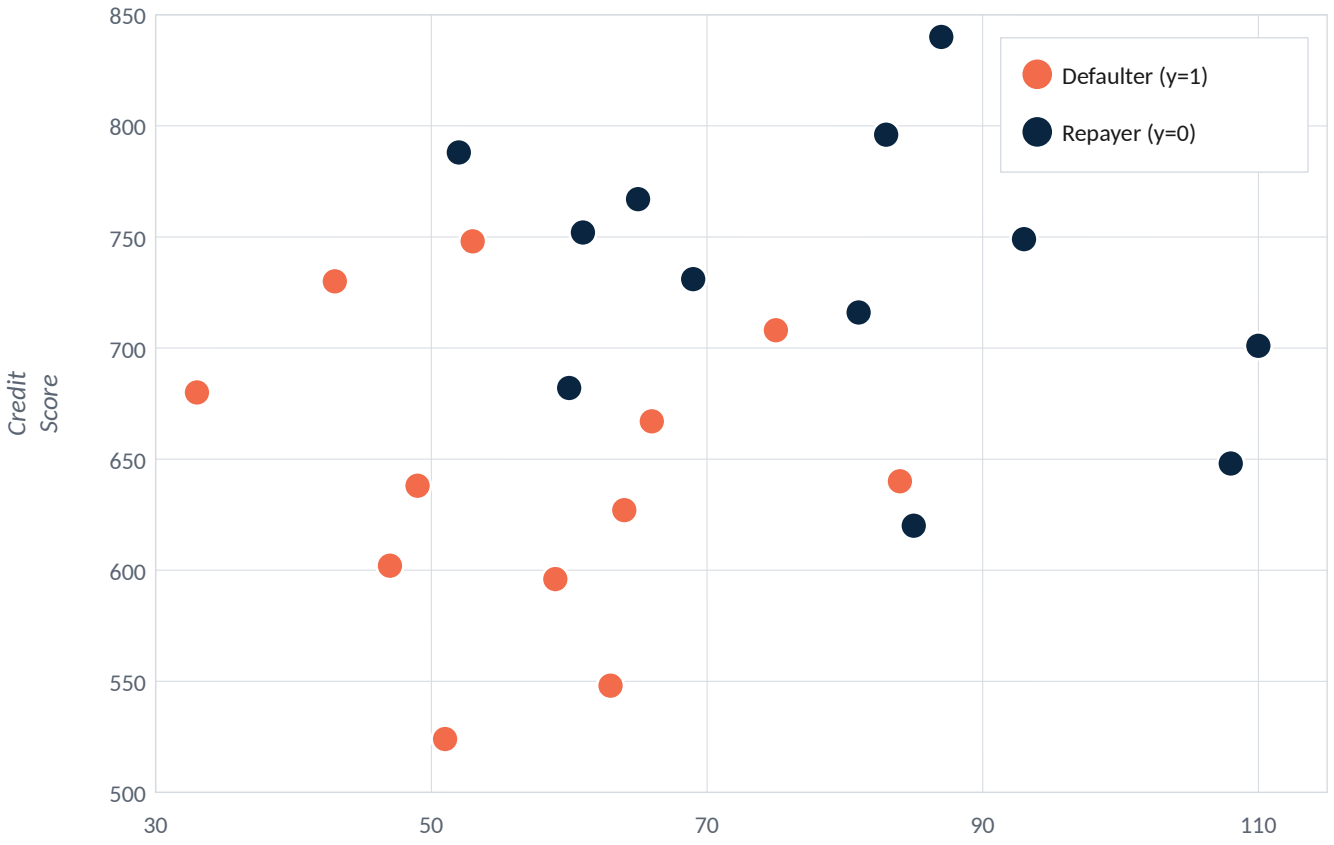
2 — Non-linear by design
No functional form assumed — the algorithm carves the feature space itself.

3 — Feature-driven splits
At each node, the algorithm asks: which feature, cut at which value, produces the purest subsets?

4 — Gini impurity is the referee
Purity is quantified. Splits happen where they most reduce mixed defaulter/repayer subsets.

The mechanics — 24 borrowers, two features

Table A-1 · 12 defaulters + 12 repayers



THE SET-UP

Target y = loan status → default (1) or repay (0)

Feature X_1 = income ($000s)

Feature X_2 = credit score

WHAT THE ALGORITHM WILL DO

Find the horizontal or vertical cut that splits these 24 dots into two boxes that are as homogeneous as possible — i.e. minimise Gini impurity.

Baseline Gini impurity: $12/24 \times (1 - 12/24) + 12/24 \times (1 - 12/24) = ½ \times ½ + ½ \times ½ = 0.5$ — *the maximum possible for a two-class problem.*

First cut — credit score = 681 wins the tournament

THE SPLIT AND WHAT LANDS WHERE

Rectangle	Repayers	Defaulters	Total
R1 · credit > 681	10	3	13
R2 · credit ≤ 681	2	9	11

THE GINI MATH

Gini(R1) = 10/13 × 3/13 + 3/13 × 10/13
= 60/169 = **0.356**

Gini(R2) = 2/11 × 9/11 + 9/11 × 2/11
= 36/121 = **0.298**

Weighted Gini(R1/R2) = 13/24 × 0.356 + 11/24 × 0.298 = **0.3286**

INFORMATION GAIN FROM THE FIRST SPLIT

BEFORE THE CUT

Gini = 0.500

(maximum impurity — 12 defaulters and 12 repayers all in one bag)

AFTER THE CUT (credit = 681)

Gini = 0.3286

Information gain = 0.500 − 0.3286 = 0.1714

This becomes the root node. The algorithm now repeats — recursively — on R1 and R2.

Recursive partitioning — the tree grows itself

CUT 2

Income = $56,500

on R1 (credit > 681)

- R3: 1 repay + 2 default → income ≤ $56,500
- R4: 9 repay + 1 default → income > $56,500

Gini → **0.2584**

CUT 3

Income = $84,500

on R2 (credit ≤ 681)

- R5: 0 repay + 9 default → pure defaulters
- R6: 2 repay + 0 default → pure repayers

Gini → **0.1306**

AND ON…

The algorithm keeps splitting

until every terminal node is pure — or the tree is stopped by a rule

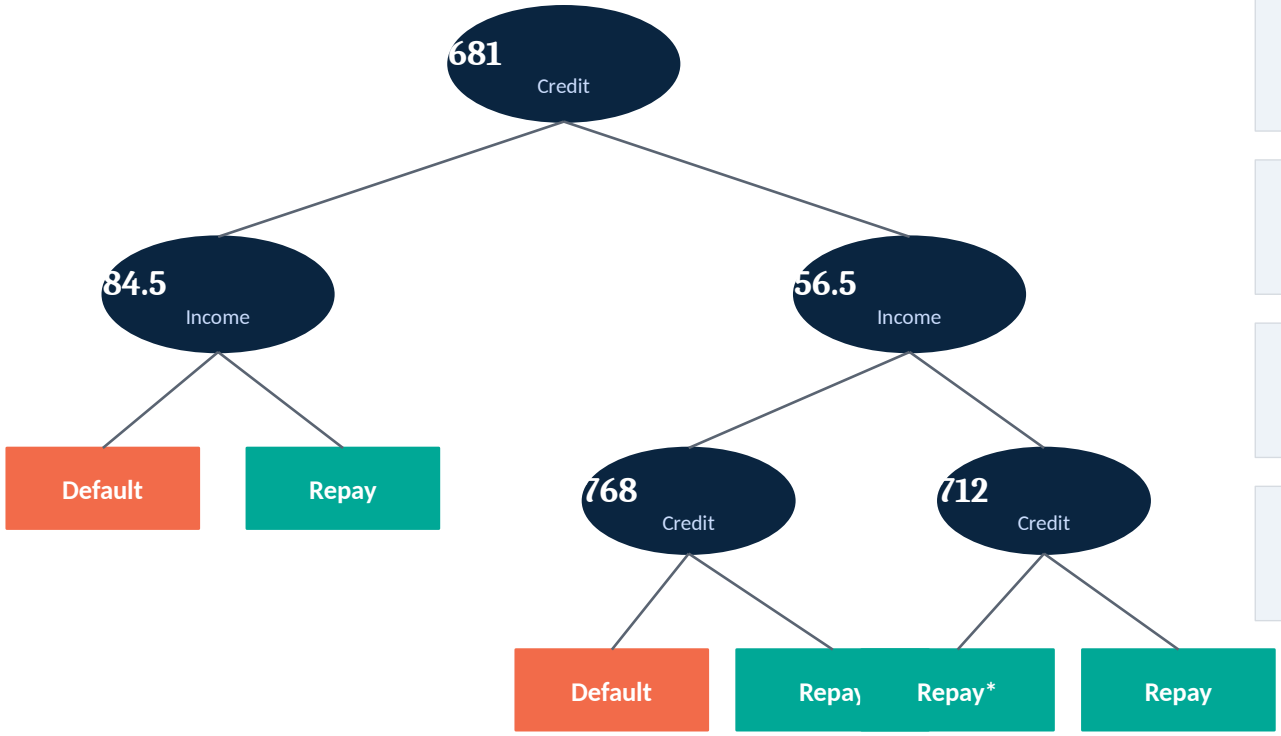
- The result is a set of rectangles, each labelled default or repay
- In our toy problem: 12 rectangles by depth 5

Gini → → **0**

Recursive partitioning in one sentence: at every step, try every possible feature-and-value combination, pick the one that reduces Gini the most, then repeat on each sub-rectangle. Left alone, the algorithm keeps going until every terminal node holds a single class.

The fully-grown tree — four if-then-else rules

STRUCTURE (SIMPLIFIED)



** One node contains 2 repayers + 1 defaulter — impurity remains.*

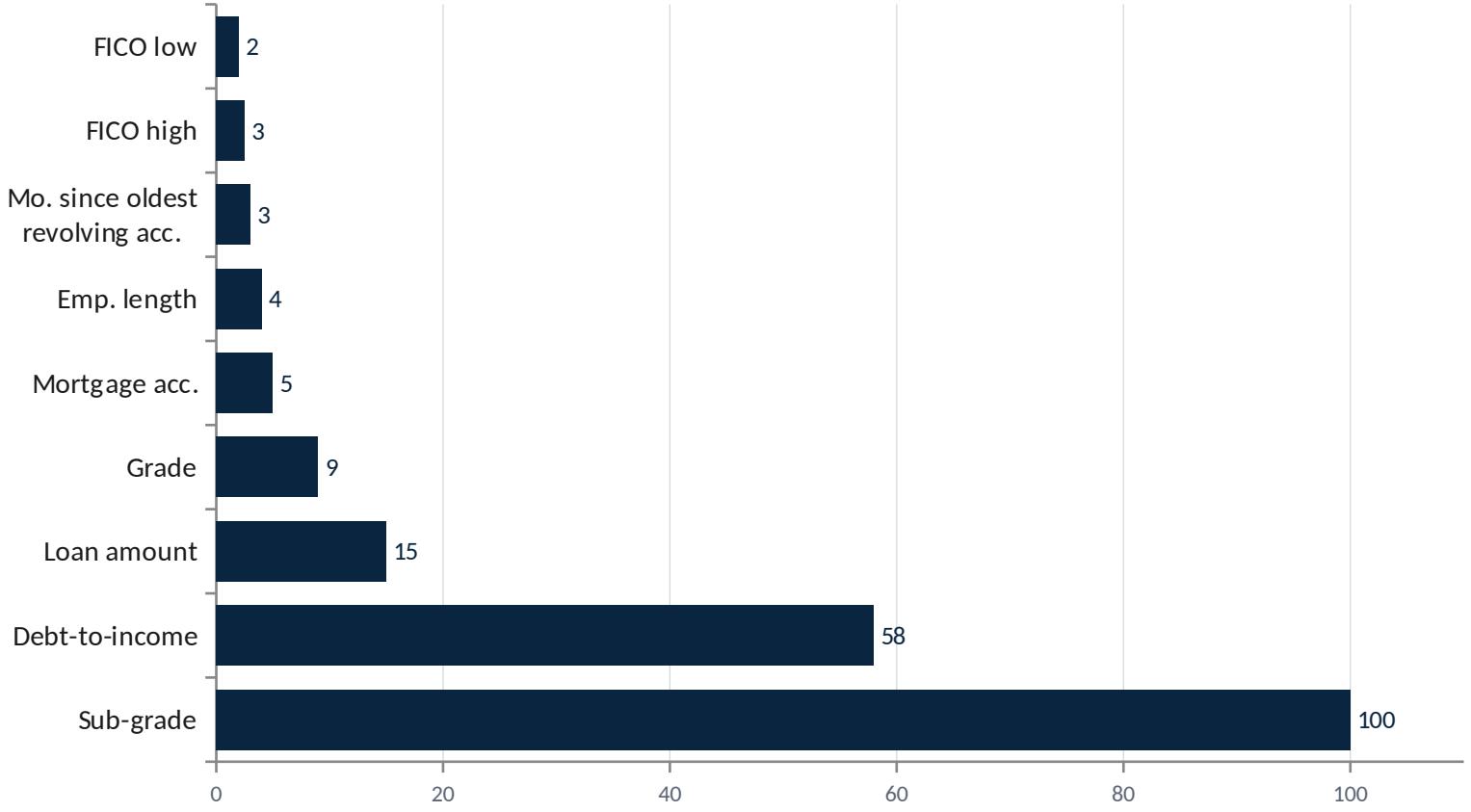
THE FOUR CLASSIFICATION RULES

1. If 681 < credit < 768 AND income < $56,500 → DEFAULT
2. If credit ≤ 681 AND income < $84,500 → DEFAULT
3. If 704.5 < credit < 712 AND income > $56,500 → DEFAULT
4. Otherwise → REPAY

Watch this quirk: Rule 3 says 'higher credit but lower income = default' — which is counter-intuitive. That's a hint the tree has overfit noise in this training subset.

Feature impact — sub-grade dominates the tree

Feature impact on the fully-grown decision tree



WHAT THIS TELLS US

Only **9 of ~70 features** register any real impact.

Sub-grade + DTI + loan amount = **~85% of the total signal.**

That's efficient — but also fragile. A single dominant feature makes the tree brittle to small data changes.

FIGEL'S PUZZLE

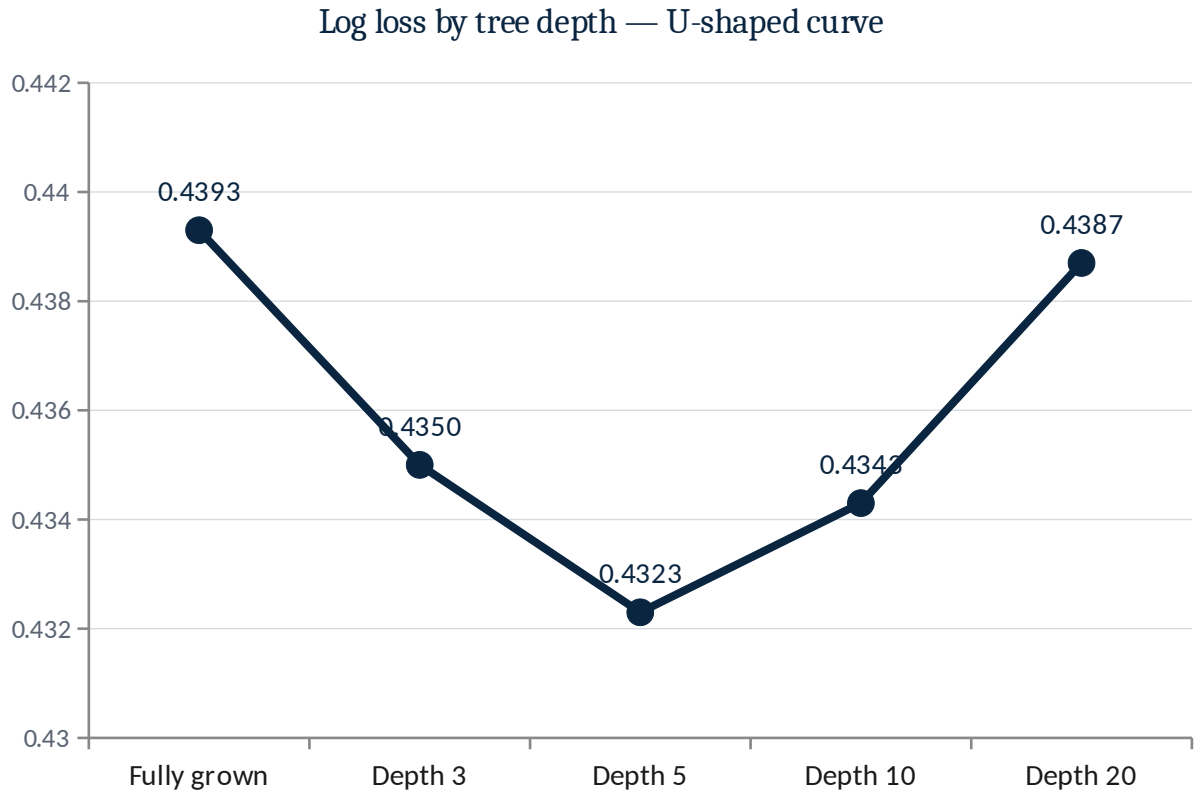
Sub-grade already encodes DTI, loan amount, FICO. So what's it 'picking up beyond' what LC's own grading captures?

The overfitting problem — and the pruning cure

WHY A DEEP TREE IS DANGEROUS

- A fully-grown tree memorises the training data — including the noise.
- Terminal nodes with 12 borrowers get their own rules. These rules almost never generalise.
- Siegel's phrase for it: 'induction bias' — assumptions latent in a particular sample that aren't true of the population.

The fix is a hyperparameter — max tree depth — chosen by cross-validation, not by looking at the training log loss.



Winner: Depth 5 · log loss = **0.4323**

Deeper trees overfit; shallower trees underfit.

Pruning reduced log loss from 0.4393 → 0.4323 — modest, but the real win is a simpler, more generalisable model.

Proof — pruning wins on data the tree has never seen

Income	Credit	Actual	Full tree pred	Pruned pred	Prob.
$60,000	680	Repay	Default*	Default*	1
$67,000	710	Repay	Default*	Repay ✓	2/3
$56,000	770	Default	Repay*	Repay*	1
$61,000	702	Repay	Repay ✓	Repay ✓	2/3
$58,000	715	Repay	Repay ✓	Repay ✓	1
Correct rate			**2 / 5**	**3 / 5**	

THE HEADLINE

+50%

more correct on new data — pruning turns a 2/5 tree into a 3/5 tree.

Why less is more: the pruned tree ignores idiosyncratic patterns near the tree's tails and generalises better.

Random Forest — many weak trees, one strong vote

PROVOST & FAWCETT'S ANALOGY

*"Imagine each model as an **expert** on the same question. If every expert had the exact same knowledge, they'd predict the same outcome and additional experts add no value.*

*Because each expert has a **slightly different perspective**, collectively they provide more information than any single expert on her own."*

HOW WE CREATE THE VARIATION

1

Bootstrapping

Each tree sees a different random sample (with replacement) of the 264K observations. Some borrowers appear multiple times, others not at all.

2

Random feature selection

At every node, the algorithm considers only $\sqrt{p} \approx \sqrt{70} \approx 8$ randomly-chosen features — not all of them. No single feature can dominate every tree.

TABLE 3 · The 8 features randomly picked at node 1 of tree 1 (√70 ≈ 8)

Purpose	Total credit lines	Employment length	Avg. current balance
FICO range-high	Credit inquiries (12m)	Revolving balance	Verification status

Tuning the forest — 250 trees × 40% features wins

THE HYPERPARAMETER GRID

HP #1 · Number of bootstrapped-sample trees

150	200	**250**	300
trees	trees	trees	trees

HP #2 · % of features considered at each node

10%	20%	30%	**40%**
features	features	features	features

THE WINNING COMBINATION (BY CV)

250 × 40%

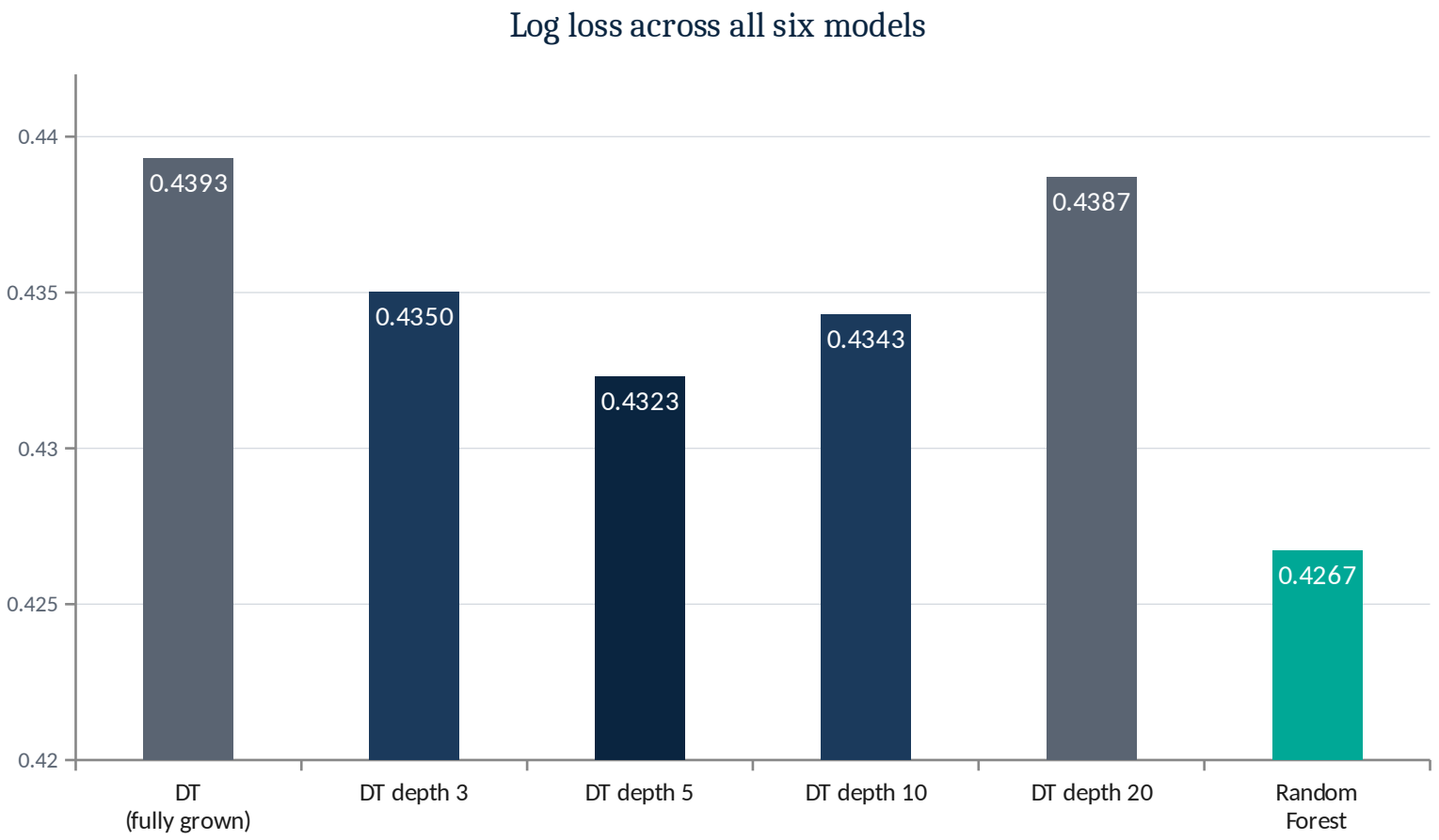
250 bootstrapped trees, each node picks from 40% of the 70 features (≈ 28 features per node — much more than √70)

FIGEL'S PUZZLE

The whole point of random-feature selection is to ***limit*** which features each node can see. Why did the optimiser pick **40%** instead of the √p rule-of-thumb (11%)?

Likely answer: LendingClub features are highly correlated. Too much suppression starves the tree of useful signal.

The showdown — Random Forest wins on log loss



WINNER

Random Forest

0.4267

log loss on 5-fold CV

Beats the fully-grown tree by 0.0126 and the best pruned tree (depth 5) by 0.0056.

READING THE SHAPE

- Pruning shape: U-curve with sweet spot at depth 5.
- RF beats every single tree — with weak trees, not strong ones.
- 0.4267 is Figel's best log loss — but still not zero.

Why RF wins — it spreads impact across more features

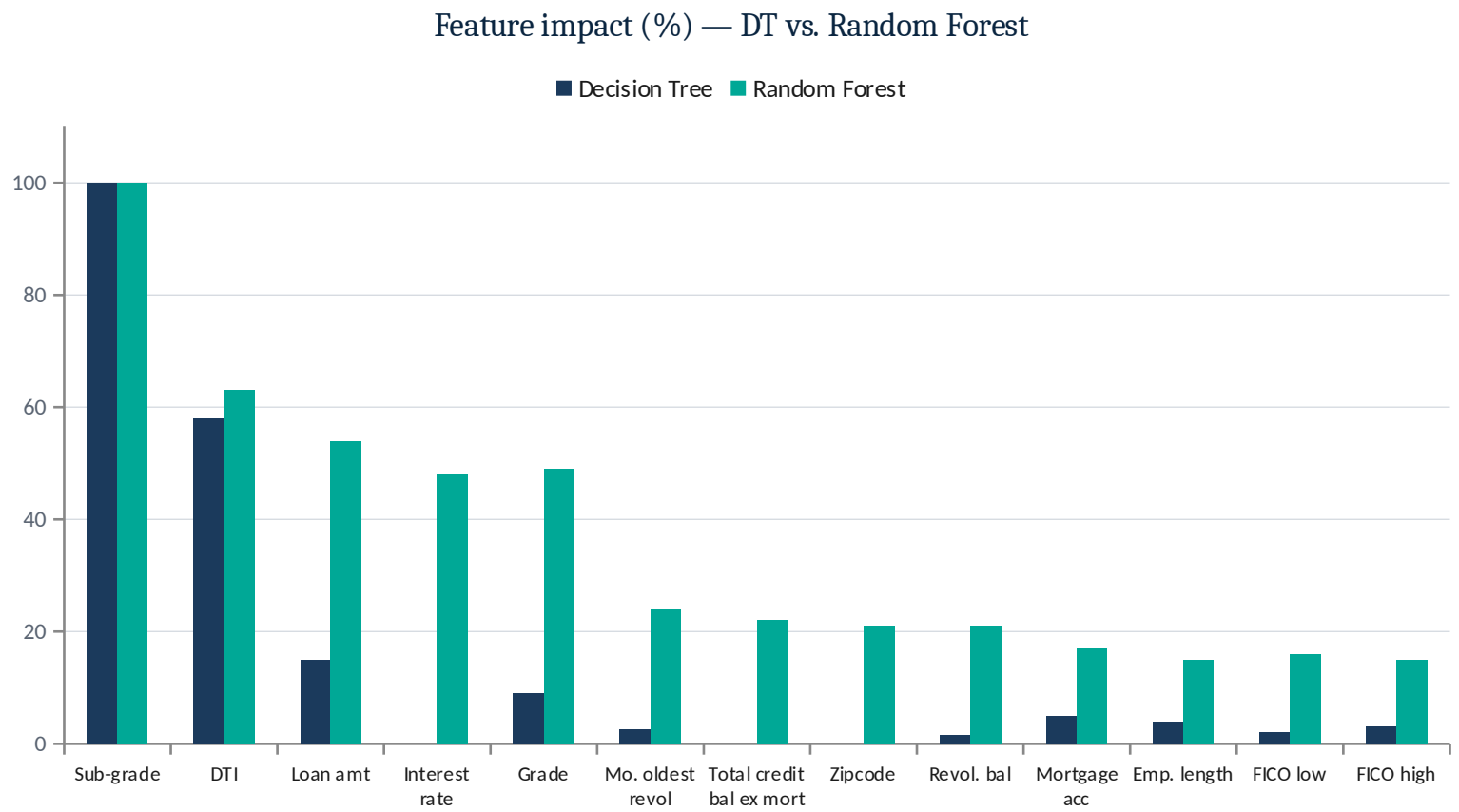


EXHIBIT 2 · IN NUMBERS

In the **Random Forest,** only **5 features (of 55)** have impact below 5%.

In the **Decision Tree,** only **4 features (of 55)** have impact above 5%.

WHAT THAT MEANS

The tree leans on a handful of dominant features — sub-grade, DTI, loan amount.

The forest recruits **10+ features** that the tree ignored — zipcode, revolving balance, interest rate — turning weak signals into a stronger vote.

Six things to take away from Case B

01 Interpretability is a feature

The decision tree's rules are readable by a loan officer, an auditor, and a regulator. That has value even when a black-box model scores slightly better.

02 Gini impurity is just accounting

Every split is chosen because it maximally reduces the weighted probability of misclassifying a randomly-picked observation.

03 Depth is the only knob that matters

A fully-grown tree memorises; a stump underfits. The U-curve of log loss vs depth localised Figel's sweet spot at depth 5.

04 Pruning proved itself on new data

5-borrower test set: full tree 2/5, pruned 3/5. Generalisation beats fitting the training data every time.

05 Ensembles beat single models

Random Forest turns 250 mediocre trees into a strong classifier — 0.4267 vs the best tree's 0.4323.

06 RF's power is diversification

Instead of relying on 4 dominant features, RF integrates signal from 10+ features that a single tree ignores. Same data, more insight.

The full arc — what Figel actually learned

A	**Data setup**	413K loans + 100+ features · target = default(1)/repay(0)
A	**First model attempt**	Perfect accuracy → discovered post-outcome leakage
A	**Feature audit (Exhibit 9)**	Removed 20 features that only exist after loan runs
A	**Second attempt**	Realistic probabilities — 16/20 correct at 50% cut-off
B	**Decision Tree**	Interpretable rules · log loss 0.4393 (unpruned)
B	**Pruning to depth 5**	log loss 0.4323 · +50% accuracy on new-data test
B	**Random Forest**	Best log loss 0.4267 · 250 trees × 40% features
B	**Feature Impact review**	RF activates 10+ features vs. DT's 4 — diversified signal

One line summary: the model can quantify borrower risk — but not the platform, the transparency, or the counterparty risks around it.

My recommendation to Figel's wealth-management firm

VERDICT

Selective, capped exposure — yes. *Whole-portfolio bet — no.*

1 Use the RF model as a screen, not the sole signal

Pair the 0.4267-log-loss random forest with LendingClub's own grade + a manual credit committee review before funding.

2 Cap the allocation at 3–5% of client AUM initially

Consistent with the return profile (~7% net) and the entity + transparency risks the model can't quantify.

3 Skew to Grade B–C, avoid E and FG

Table 3's net returns are similar (7–8%) but default rates diverge sharply — B/C offer the same reward with less tail risk.

4 Re-underwrite quarterly

Income-verification rates dropped from 49% → 26.8% in 3 years. Any drift in LC's own controls (Jefferies-style) should trigger a stop-loss.

Open for discussion: *would you build the same model in India — and which features would break first?*

QUESTIONS & DISCUSSION

Thank you.

PRESENTED BY

Kishore Kunal

Roll No. 38BM21 · PGDM-BM & PGDM-BA · MDI Gurgaon

COURSE & INSTRUCTOR

Strategy Formulation & Implementation

Prof. Sangeeta Shah Bharadwaj · Term IV (Jun–Aug 2026)